

Juan Sebastián Numpaque Roa

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EDUCATION

Colegio Corazonista H.H del Sagrado Corazón, Bogotá D.C, Colombia

- High school diploma. Jan 2001 – Dec 2012

Universidad de los Andes, Bogotá D.C, Colombia

- Bachelor of Science (B.Sc.) in Mathematics. Thesis : “The Geometry of Gerbes over Spheres”. Advisor: Dr. Alexander Cardona. GPA: 3.9/5.0. Jan 2014 – Dec 2018
- Bachelor of Science (B.Sc.) in Systems and Computing Engineering. Thesis : “Optimizing Work Stealing Algorithms”. Advisor: Dr. Nicolás Cardozo. GPA: 3.9/5.0. Jun 2014 – Dec 2020
- Master of Science (M.Sc.) in Mathematics. Thesis : “The Spectral Description of the Hitchin Fibration”. Advisor: Dr. Florent Schaffhauser. GPA: 4.78/5.0. Jan 2019 – Aug 2021

Sorbonne Université, Paris, France

- Master 2 (M.Sc.) in Mathématiques Fondamentales. Thesis: “Counting absolutely indecomposable quiver representations”. Advisors: Dr. Emmanuel Letellier and Dr. Florent Schaffhauser. GPA: 15/20 (Mention Bien). Sep 2021- Sep 2022

Universidade do Porto, Porto, Portugal

- Doctor of Philosophy (Ph.D.) in Mathematics. Advisor: Dr. Peter Gothen Oct 2022- Ongoing

DISTINCTIONS & AWARDS

- Fourth best national score in SABER 11 test (Colombia’s SAT analogue). Apr 2013
- “Alberto Magno” award. Granted by Universidad de los Andes to the best admission scores for the first academic semester of 2013. Universidad de los Andes. Bogotá D.C. Apr 2013
- “Bachilleres por Colombia- Mario Galán Gómez” full scholarship. Granted by Ecopetrol (Colombia’s national oil company) to the best scores in SABER 11 test. Bogotá D.C. Aug 2013
- Full scholarship awarded by the math department of Universidad de los Andes to the best scores in the admission test to the Mathematics M.Sc. program. Universidad de los Andes. Bogotá D.C. Dec 2018

PUBLICATIONS

- Numpaque-Roa, J.S., Cardozo, N.(2022). [Evaluation of Work Stealing Algorithms](#), in Gonzalez, E., Curiel, M., Moreno, A., Carrillo-Ramos, A., Páez, R., Flórez-Valencia, L. (eds) Advances in Computing. CCC 2021. Communications in Computer and Information Science, vol 1594. Springer, Cham.
- Numpaque-Roa, J.S., Schaffhauser, F.(2025). [Flat torsors in complex geometry](#), in P.L. del Ángel R., F. Neumann, and A.H.W. Schmitt (eds.) Moduli, Motives and Bundles: New Trends in Algebraic Geometry. Cambridge: Cambridge University Press (London Mathematical Society Lecture Note Series), pp. 106–142.
- Numpaque-Roa, J.S.(2025). [Tensor products of quiver bundles](#). The Quarterly Journal of Mathematics, Volume 76, Issue 4, December 2025, pp. 1235–1266.

COURSES, SCHOOLS & CONFERENCES

- V Encuentro Colombiano de Combinatoria**, Universidad Nacional y Universidad de Antioquia. Medellín, Antioquia. Jun 2016
- Complex Foliations, Dynamics and Geometry**, Universidad Federal Fluminense. Niterói, Rio de Janeiro, Brasil. Jul 2018
- Summer Research Institute - Geometry and Theoretical Physics**, Villa de Leyva, Boyacá, Colombia.
 - Contributed talk: “From Line Bundles to Gerbes”. Jul 2019
- School of Poisson Geometry**, IMPA, Rio de Janeiro, Rio de Janeiro, Brasil. Dec 2019
- International Conference on Poisson Geometry**, IMPA, Rio de Janeiro, Rio de Janeiro, Brasil. Dec 2019
- Geometric and Analytic aspects of moduli spaces of Higgs bundles**, Université de Strasbourg, Strasbourg, France. Jun 2021
- Short course on Nonabelian Hodge theory**, Waseda University, Tokyo, Japan. Jul 2021
- Geometric Structures (re)United**, University of Illinois at Chicago, Chicago, United States of America. Jun 2022
- Young Algebraic Geometer Seminar**, Universidad de los Andes, Bogotá D.C.
 - Contributed talk: “Counting absolutely indecomposable quiver representations”. Sep 2022
- Forschungsseminar AG-Komplexe Analysis - Prof. Dr. Alexander Schmitt**, Freie Universität, Berlin, Germany.
 - Contributed talk: “The spectral description of the Hitchin Fibration”. Nov 2022
- Geometry Seminar**, Universidade de Coimbra, Coimbra, Portugal.
 - Contributed talk: “A gentle introduction to the Hitchin Fibration”. Feb 2023
- Higgs bundles, character varieties and higher Teichmüller spaces**, ICMAT, Madrid, Spain. Apr 2023
- Workshop and school on Complex Lagrangians, Integrable Systems, and Quantization**, University of Oxford, United Kingdom. Jun 2023
- Vector bundles and combinatorial algebraic geometry**, Goethe University Frankfurt am Main, Frankfurt, Germany.
 - Contributed Mini-Talk: “From Higgs to Quiver bundles”. Oct 2023
- Moduli spaces of vector bundles and curves**, Université de Cergy-Pontoise, Paris, France. Dec 2023
- Arbeitsgemeinschaft: Geometry and Representation Theory around the P=W Conjecture**, Mathematisches Forschungsinstitut Oberwolfach, Oberwolfach, Germany. Apr 2024
- 2nd Simons Math Summer Workshop: Moduli**, Simons Center for Geometry and Physics, Stony Brook, United States of America. Jul 2024
- MaPhyAG Workshop: Integrability, Geometry and QFT**, Universidad Complutense de Madrid, Madrid, España. Oct 2024
- Spring School - The Dynkin Classification**, Ruhr Universität Bochum, Bochum, Germany. Apr 2025
- Stratifications of Higgs bundle moduli spaces and related topics**, Universidade de Santiago de Compostela, Santiago de Compostela, Spain. May 2025

Vector Bundles on Curves – Quivers and Sheaves , Utrecht University, Utrecht, Netherlands.

- In this workshop I contributed with a 40 minute talk on my article: “Tensor products of quiver bundles”. [Here](#) you can find the slides. May 2025

TiME 2025 Summer School on Invariant Theory, Lie-Størmer Center, Tromsø, Norway. May 2025

Geometric Aspects of the Langlands Correspondence, Universität Duisburg-Essen, Essen, Germany. Aug 2025

One-day workshop “Enumerative Geometry and Geometric Representation Theory” , Università di Pisa, Pisa, Italy.

- In this workshop I contributed with a 30 minute talk on my article: “Tensor products of quiver bundles”. Dec 2025

VBAC 2026: Curves, Surfaces and Moduli , Universidad Autónoma de Yucatán, Mérida, Mexico.

- In this workshop I will present a poster on my article: “Tensor products of quiver bundles”. [Here](#) you can find it. Jan 2026

Bridgeland Stability Doctoral School, Universidade de Santiago de Compostela, Santiago de Compostela, Spain. Mar 2026

LANGUAGES

- Spanish: Mother language.
- English: Fluent. 99/120 in TOEFL iBT test.
- Portuguese: Fluent. “Intermediário Superior” certificate in the CELPE-BRAS test.
- German: Elementary.
- French: Elementary.

WORK EXPERIENCE

Universidad de los Andes, Bogotá D.C, Colombia

- Teacher Assistant: Pre-calculus, Differential Calculus and Analysis on Manifolds.
- Graduate Teaching Assistant: Running the exercise sessions for the Linear Algebra, Differential Calculus, Integral Calculus, Multivariable Calculus and Calculus 3 and Linear Algebra (for Economists and Business Administrators) courses. Jan 2019 – Jun 2021
- Graduate Research Assistant: Participation on research project called “The basic gerbe over a compact simple Lie group”. Jun 2019 – Aug 2019
- Graduate Administrative Assistant: Producing, filming and editing video tutorials for the Multivariable Calculus course. Jun 2020 – Aug 2020

Laude, Bogotá D.C, Colombia

- Laude is an online platform that offers online resources and video tutorials in different subjects to colombian students. My role in this start-up was to produce and write video tutorials on mathematics subjects for middle school students. Jun 2021 – Sep 2021

PROGRAMMING LANGUAGES

$\LaTeX$, MATLAB, R, JAVA, Python.

REFERENCES

- Prof. Peter Gothen (pbgothen@fc.up.pt)
- Prof. Florent Schaffhauser (fschaffhauser@mathi.uni-heidelberg.de)